

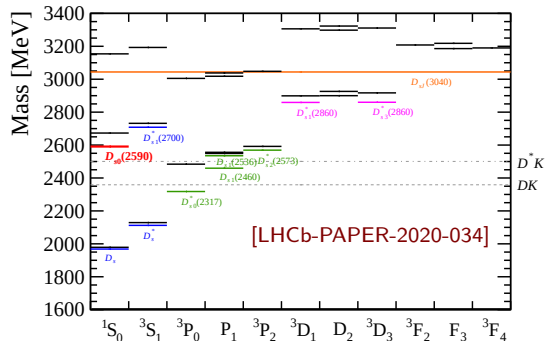
Ω_c^{**0} and other charm baryons

Mikhail Mikhasenko
CERN, Switzerland
(soon ORIGINS Cluster, Munich)

June 11st, 2021
Heavy-flavor lunch

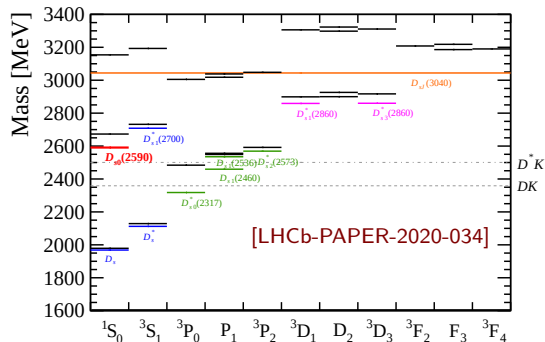
Why studying charm hadrons?

- Two types of exotics:
 - ▶ Genuine exotics – no convent. expected
 - ▶ **Mixture** – the conventional states are influenced particle interaction. Mass shift towards the strongly-coupled threshold.
- b/c hadrons are narrow
- Level-splitting hierarchy due to $1/m_Q$



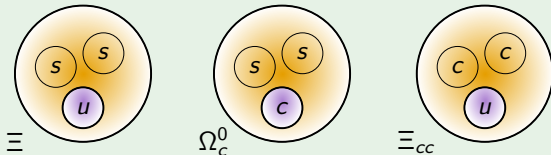
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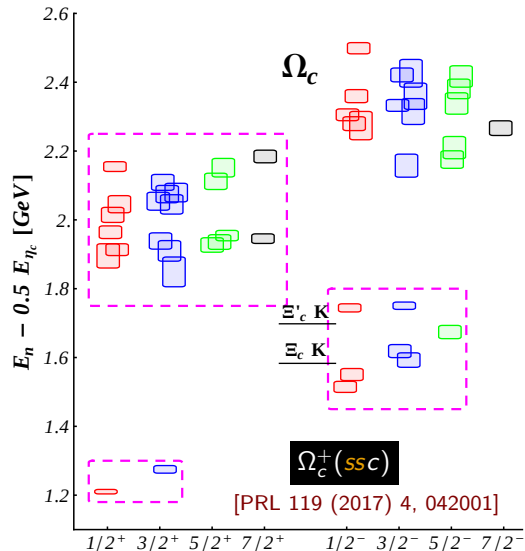
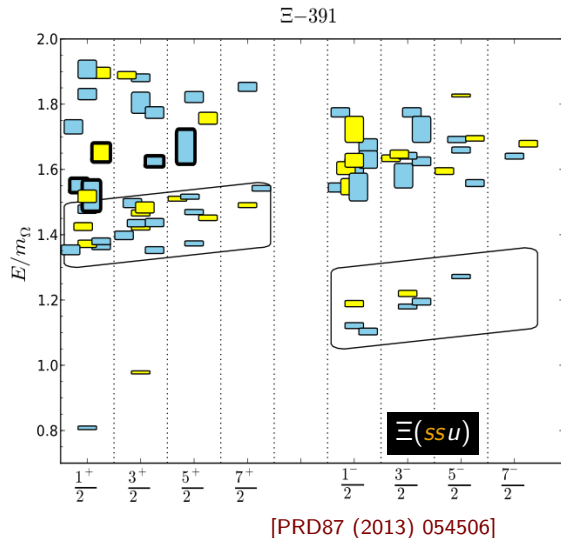


Baryons

- Quark-diquark picture
- Many deeply connected sectors, e.g.

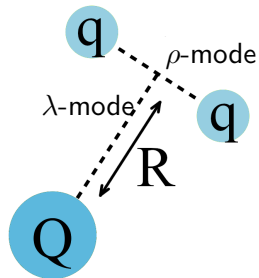


Lattice QCD: 5 λ modes and 2 ρ modes

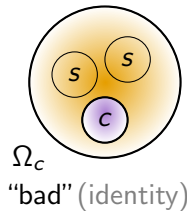
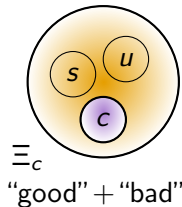
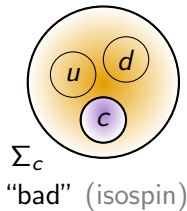
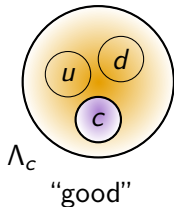


Heavy-quark-diquark system

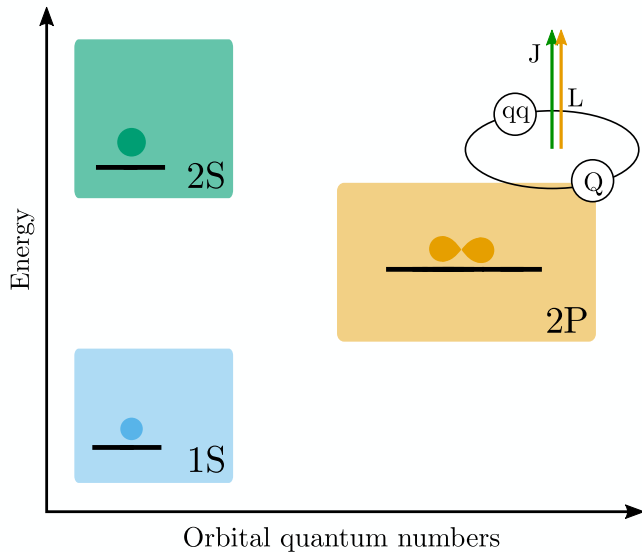
Charm-baryon sectors



- Heavy quark is **static** and **spinless** in the limit $m_Q \rightarrow \infty$.
- Excitations of Qqq are governed by the light diquark
- $q \uparrow^{(J^P=\frac{1}{2}^+)} \otimes q \uparrow^{(J^P=\frac{1}{2}^+)} \Rightarrow \underbrace{\uparrow\downarrow^{(J^P=0^+)}}_{\text{“good”}} \text{ and } \underbrace{\uparrow\uparrow^{(J^P=1^+)}}_{\text{“bad”}}$
- Excitation pattern is different for “good” and “bad” diquarks



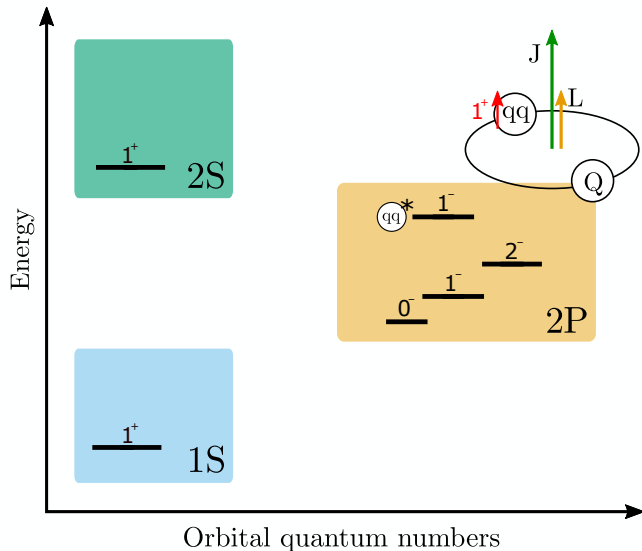
Excitation spectrum of baryons with “bad” diquark



Structure:

- Radial and orbital excitations

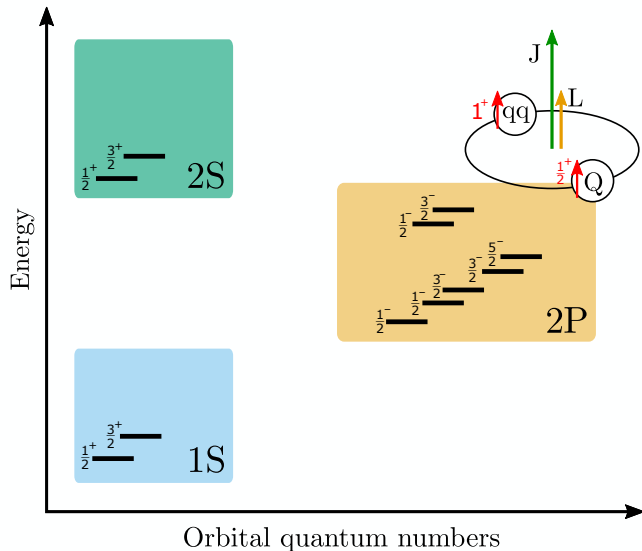
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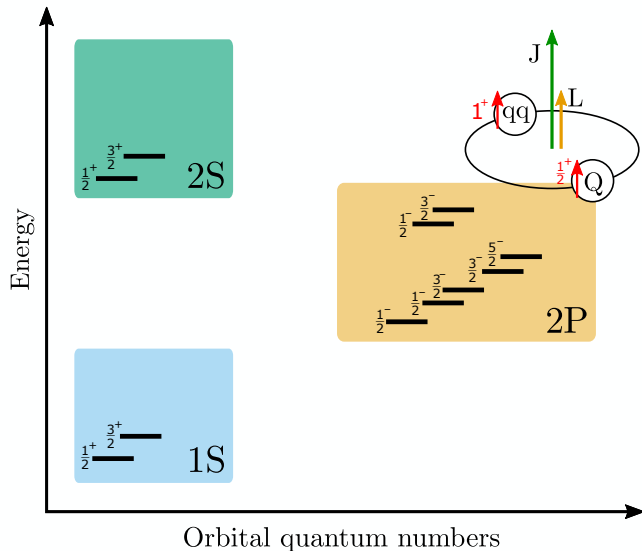
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Excitation spectrum of baryons with “bad” diquark



Structure:

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Counting of states

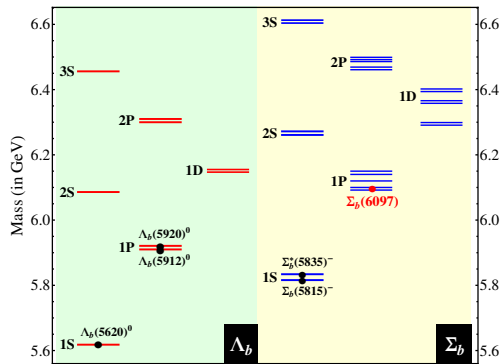
– model independent

Size of splitting, the order

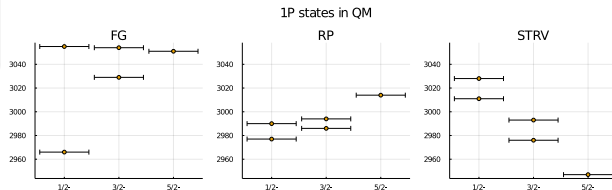
– differs from model to model

Phenomenological models

- Agree on the general pattern
- Agree on relation between the sectors



[Bing Chen et al., PRD98 (2018) 074032]



[Faustov-Galkin, EPJ Web Conf. 204 (2019) 08001]

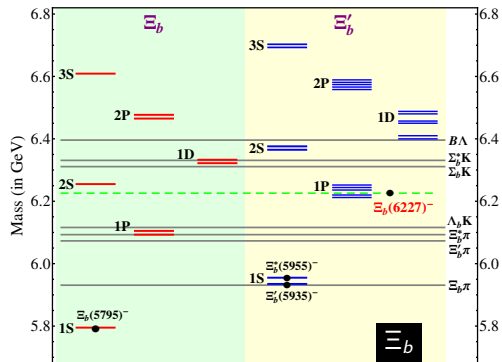
[Roberts-Pervin, Int.J.Mod.Phys.A 23 (2008) 2817-2860]

[Shah-Thakkar-Rai-Vinodkumar, EPJA 52 (2016) 10, 313]

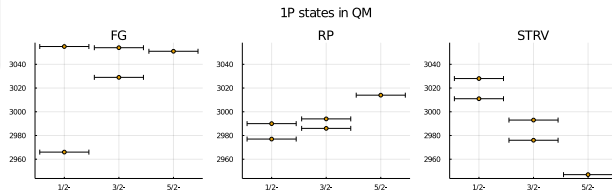
- Often neglect the diquark excitation
- Disagree on the splitting and the order

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[Bing Chen et al., PRD98 (2018) 031502]



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- Often neglect the diquark excitation
- Disagree on the splitting and the order

Strong transitions to a baryon and a pseudoscalar

$$\Omega_c^{**0} \rightarrow \Xi_c^+ K^-$$

$$\Xi_c^{**0} \rightarrow \Lambda_c^+ K^-$$

$$\Sigma_c^{**0} \rightarrow \Lambda_c^+ \pi^-$$

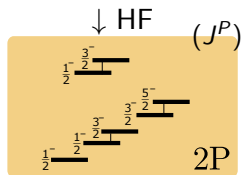
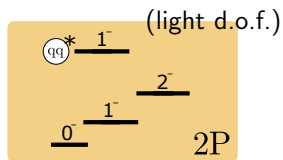
$$\Omega_b^{**-} \rightarrow \Xi_b^0 K^-$$

$$\Xi_b^{**-} \rightarrow \Lambda_b^0 K^-$$

$$\Sigma_b^{**-} \rightarrow \Lambda_b^0 \pi^-$$

HQSS ($m_Q \rightarrow \infty$) gives the selection rule based on the light d.o.f. [Chiladze, Falk, PRD 56 (1997)]

Can be applied to the decay to ground-state baryon and pseudoscalar ($B + P$):



J^P (light d.o.f.)	J^P	$B + P$ (light d.o.f.)	$B + P$
0 ⁻	1/2 ⁻	S-wave	S-wave
1 ⁻	1/2 ⁻	forbidden	S-wave
	3/2 ⁻	forbidden	D-wave
2 ⁻	3/2 ⁻	D-wave	D-wave
	5/2 ⁻	D-wave	D-wave
1 ⁻	1/2 ⁻	forbidden	S-wave
	3/2 ⁻	forbidden	S-wave

Four of seven are expected to be suppressed

Experimental observations



LHCb

beauty

[display]

Ring Imaging CHerenkov

Kaon ID $\sim 95\%$
 π/K mis ID $\sim 5\%$

Dipole Magnet

Hadronic
CALorimeter

Muon
Stations

Muon ID $\sim 97\%$
 π/μ mis ID $\sim 1-3\%$

Vertex LOcator

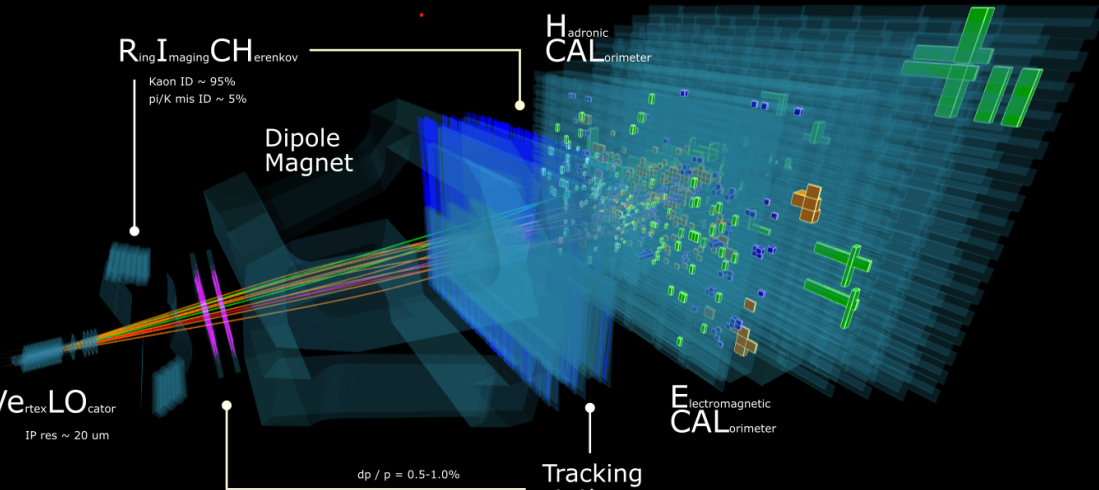
IP res $\sim 20 \mu\text{m}$

$dp/p = 0.5-1.0\%$

Tracking
stations

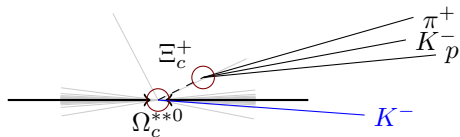
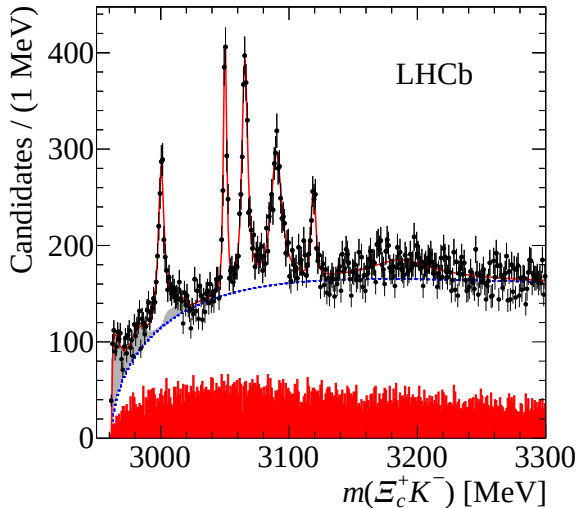
Electromagnetic
CALorimeter

pp collider (7+7 TeV)



Ω_c^{**0} states in prompt production

[LHCb, PRL 118, 182001 (2017)]



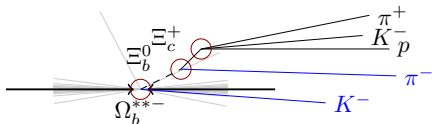
- 5 super-narrow structures
- 1 broad structure
- 3 gray components partially reconstructed
 $\Omega_c^{**0} \rightarrow \Xi_c'^+ (\rightarrow \Xi_c^+ \gamma) K$

A popular J^P assignment:
 the narrow states are λ modes
 in the natural order $\frac{1}{2}^-, \frac{1}{2}^-, \frac{3}{2}^-, \frac{3}{2}^-, \frac{5}{2}^-$.

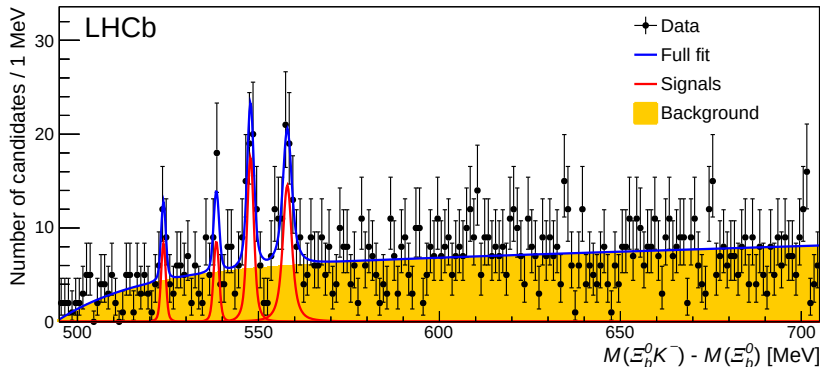
[Karliner:2017kfm, Padmanath:2017Ing, Wang:2017zjw]

$\Omega_b^{*-}-$ states in prompt production

[LHCb, PRL 124, 082002 (2020)]



- $\sim 100\times$ smaller statistics
- Four peaks are seen with significance $> 3\sigma$
- No fifth narrow state
- Do we see HQSS suppression for the first two?



The new states are

$$\Omega_b(6315)^-$$

$$\Omega_b(6330)^-$$

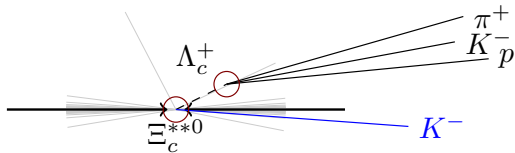
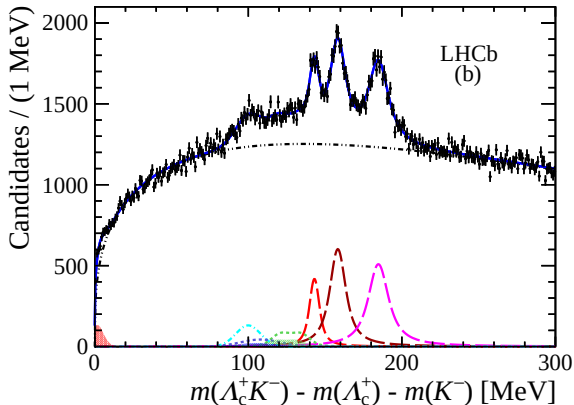
$$\Omega_b(6340)^-$$

$$\Omega_b(6350)^-$$

Ξ_c^{**0} states in prompt production

[LHCb, PRL 124, 222001 (2020)]

$$\Xi_c^{**0} \begin{pmatrix} c \\ s \\ d \end{pmatrix} \rightarrow \Lambda_c^+ \begin{pmatrix} c \\ u \\ d \end{pmatrix} + K^- \begin{pmatrix} s \\ \bar{u} \end{pmatrix}$$

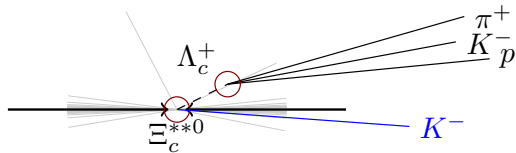
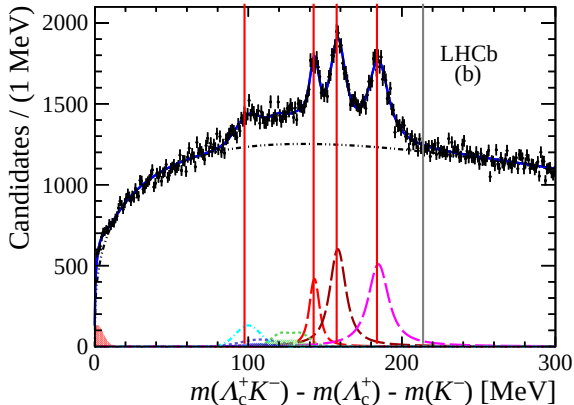


- Four structures are clearly visible
- More combersome partially-reconstructed decays
- No fifth narrow state
- The peaks are wider(!)

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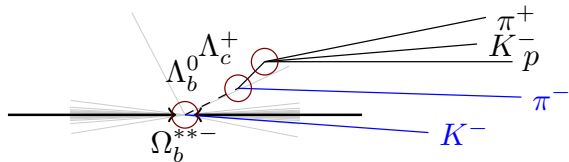
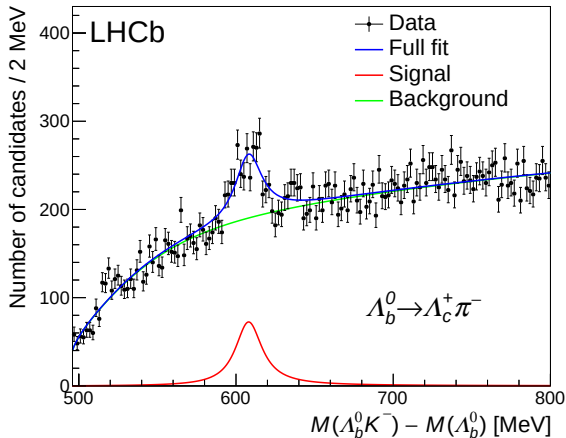


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- Same peak spacing as for Ω_c^{**0}

Ξ_b^{*-} states in prompt production

Λ_b is reconstructed in two final states: $\Lambda_c^+ \pi^-$ and $\Lambda_c^+ \pi^+ \pi^- \pi^-$

[LHCb, PRD 103, 012004 (2021)]

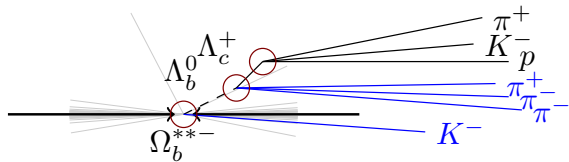
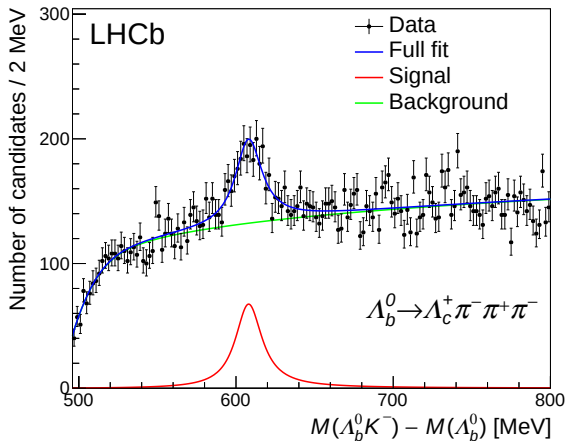


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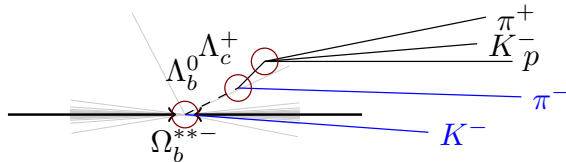
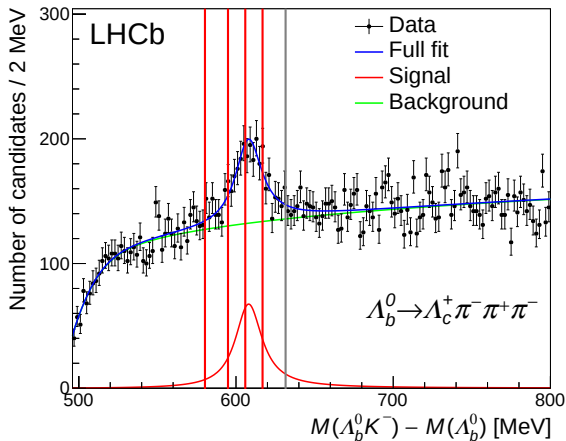


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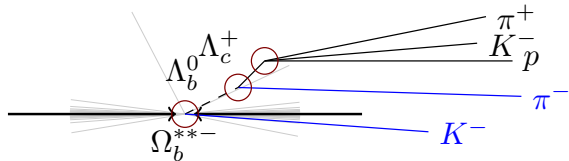
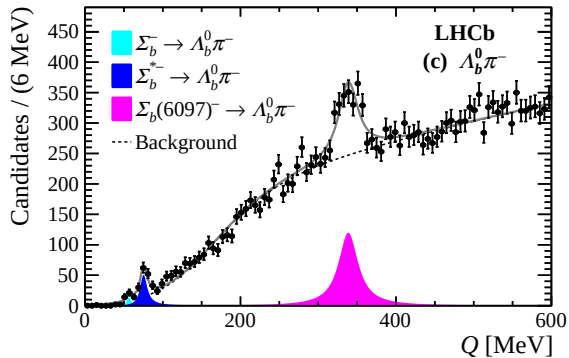


- One clear structure but anomalously broad
- Splitting from Ω_b^{*-} indicates that it will be hard to resolve.

$\Sigma_b^{*\pm}$ states in prompt production

Λ_b^0 is reconstructed in $\Lambda_c^+ \pi^-$

[LHCb, PRL 122(2019) 012001]

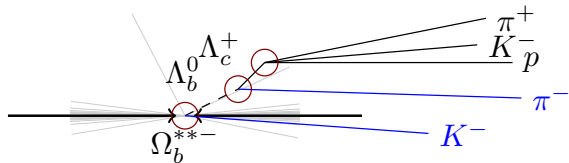
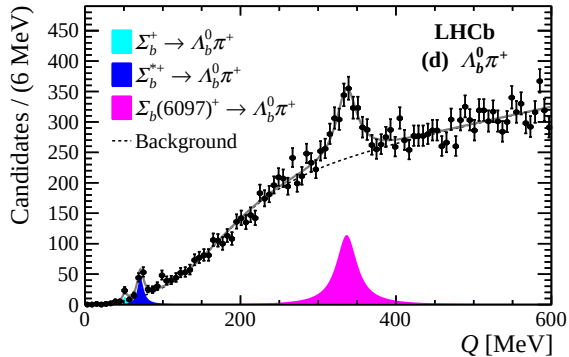


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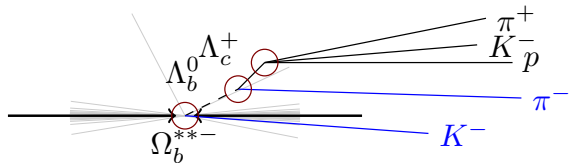
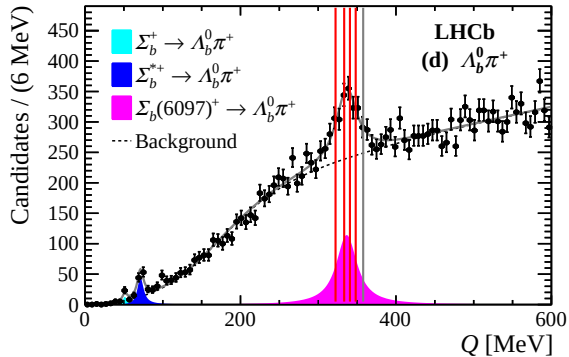


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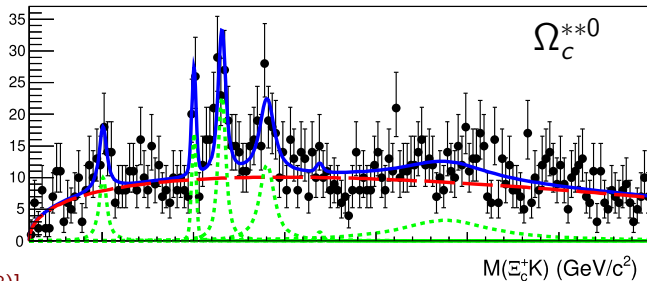
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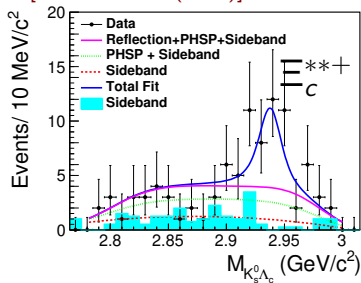
[LHCb, PRL 122(2019) 012001]



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[EPJC 78, 252 (2018)]

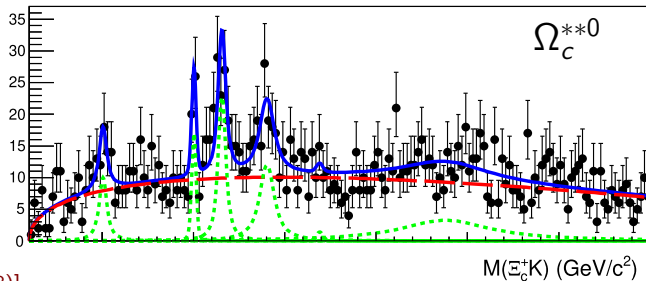


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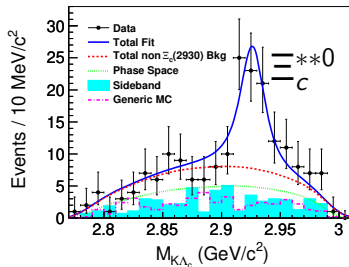
$$e^+ e^- \rightarrow \Omega_c^{*0} (\rightarrow \Xi_c^+ K^-) X$$

- Ξ_c^{*0} structures in B decay

$$B^0 \rightarrow \Lambda_c^+ \Lambda_c^- K^0$$



[EPJC 78, 928 (2018)]



- four Ω_c^{*0} structures are observed in prompt

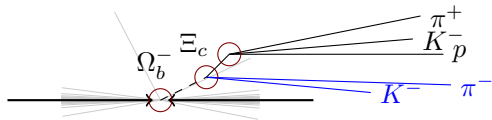
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- Ξ_c^{*0} structures in B decay

$$B^+ \rightarrow \Lambda_c^+ \Lambda_c^- K^+$$

The first exclusive observation of Ω_c^{**0}

In $\Omega_b^- \rightarrow \Xi_c^+ K^- \pi^-$ decay

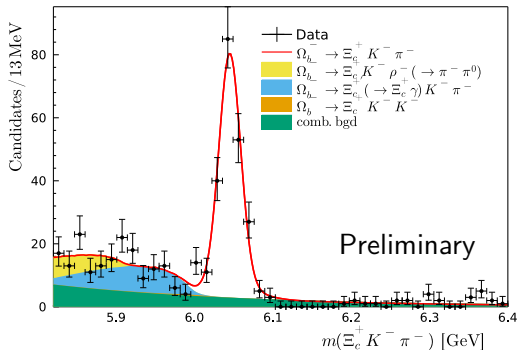
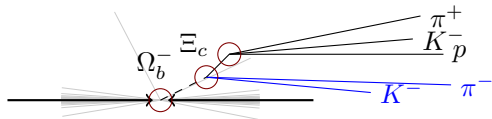


[LHCb-PAPER-2021-012, in preparation]

- Strict exclusivity cut $\Rightarrow$ No feed down!
- Same four peaks (no clear fifth)
- + **the threshold structure** (5.3σ)

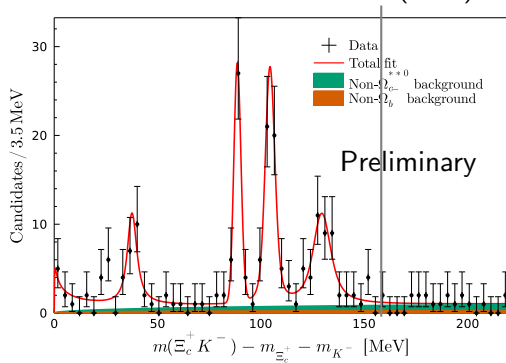
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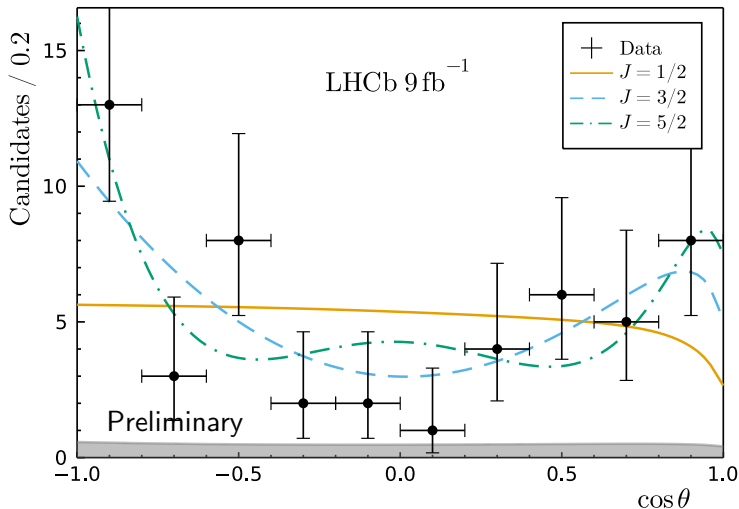
Angular analysis of $\Omega_b^- \rightarrow \Omega_c^{*0}(\rightarrow \Xi_c^+ K^-)\pi^-$

[LHCb-PAPER-2021-012, in preparation]

$\Omega_c(3065)^0$

- Spin of Ω_b^- is $1/2$
- Ω_c^{*0} cannot have spin projection $> 1/2$
- $\Rightarrow$ non-trivial angular dependence for $J = 3/2$, $J = 5/2$.
- Noticable inefficiency at $\cos\theta = 1$ (soft K^-).

3.6σ : $J(\Omega_c(3065)^0) = 1/2$
 2.2σ : $J(\Omega_c(3050)^0) = 1/2$



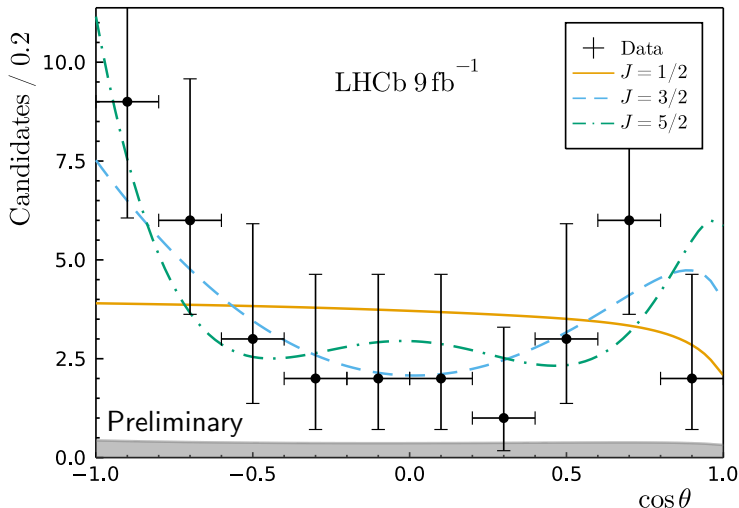
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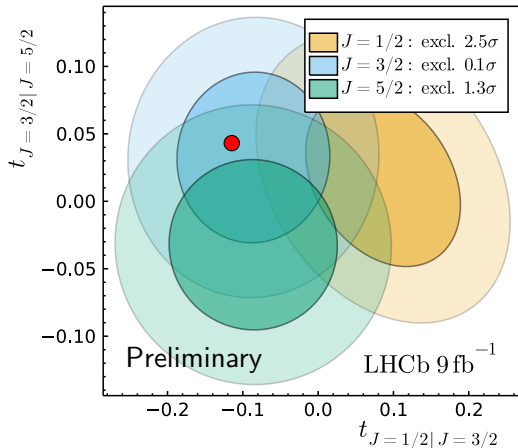


Spin hypotheses testing

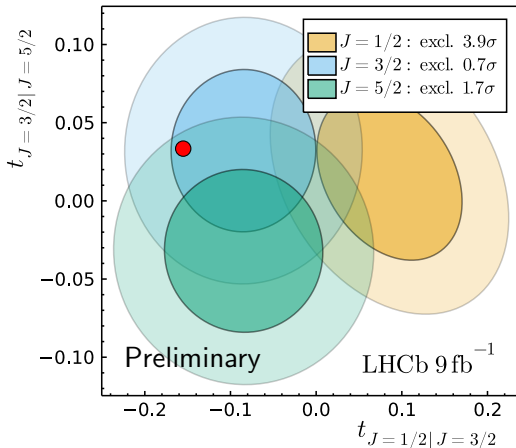
[LHCb-PAPER-2021-012, in preparation]

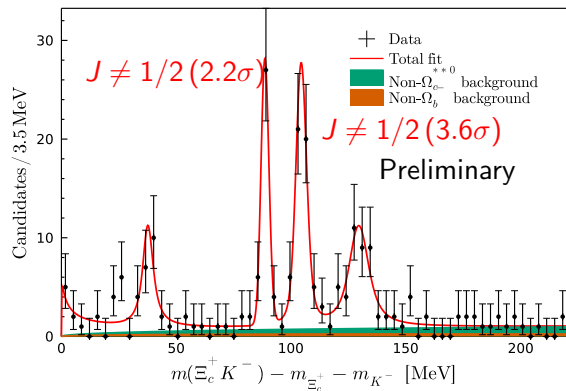
- 2d Log Likelihood ratio as a test statistics $t_{J|J'} = \frac{1}{N} \sum_{i=1}^N \log(I_J(\cos \theta_i)/I'_{J'}(\cos \theta_i))$

$\Omega_c(3050)^0$



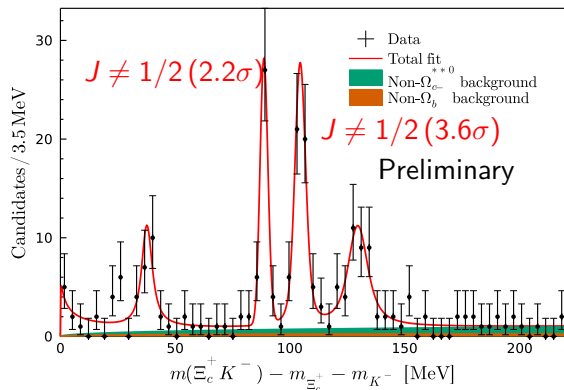
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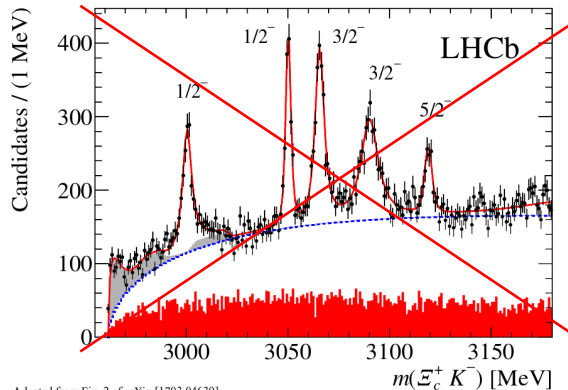


Combined spin test

[LHCb-PAPER-2021-012, in preparation]



Excluded by 3.6σ

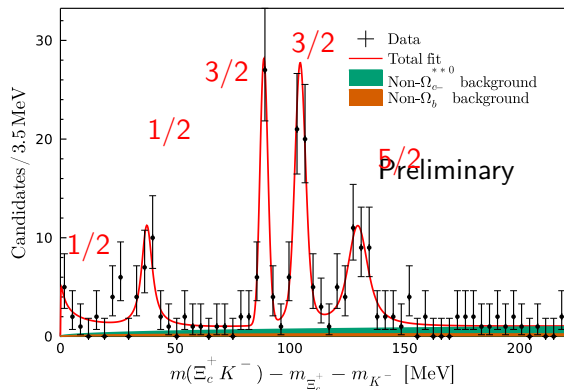


Adapted from Fig. 2 of arXiv:[1703.04639]

to be revisited!

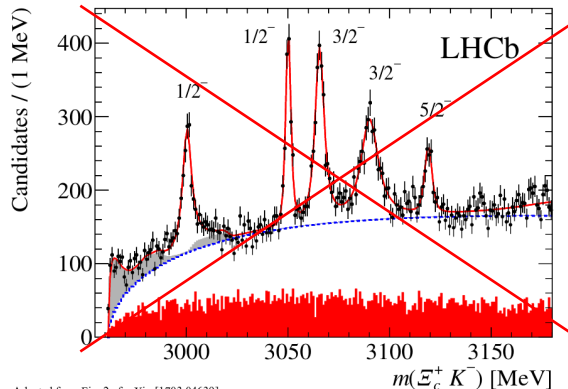
Combined spin test

[LHCb-PAPER-2021-012, in preparation]



One plausible assignments

Excluded by 3.6σ

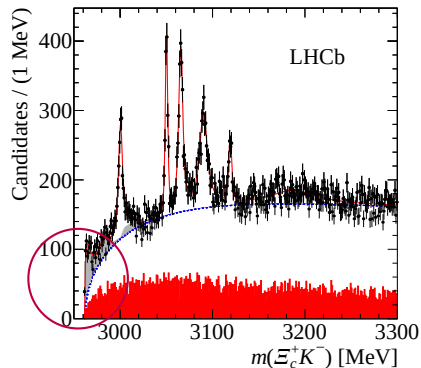


Adapted from Fig. 2 of arXiv:[1703.04639]

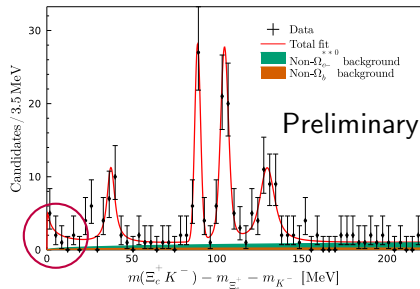
to be revisited!

The threshold structure

[LHCb-PAPER-2021-012, in preparation]



- Explained in the prompt analysis by the partially reconstructed $\Omega_c(3065)^+ \rightarrow \Xi_c'^+ K^-$ with anomalously large coupling.



- Exclusive analysis: no feed down is possible
- Other non-physical sources are excluded
- Significance in the nominal fit is 5.3σ , 4.3σ including systematics
- No model sensitivity due to the low statistics

Further investigation is needed!

Summary

$1P$ multiplet of “bad” diquark – a fundamental piece of the baryon spectroscopy puzzle

- 7 states: five λ -modes and two ρ -modes
- In common for Ξ , $\Sigma_{b/c}$, $\Omega_{b/c}$, $\Xi_{b/c}$, Ξ_{cc}
- State splittings indicate spin-orbit and “hyperfine” interaction
- Charm (beauty) baryons are narrow – good chance to resolve it:
 - ▶ Assign quantum numbers, reveal the light quark dynamics
 - ▶ Identify diquark excitation for the first time

The threshold structure

- Is a state? (a partner of $D_s(2317)$)
Compact / Molecular component?
- Present at other sectors?

The fifth narrow state

- Why only seen in inclusive $\Xi_c^+ K^-$
- The first manifestation of the diquark excitation? $2S$ states?